



Tubender

Final Report Document

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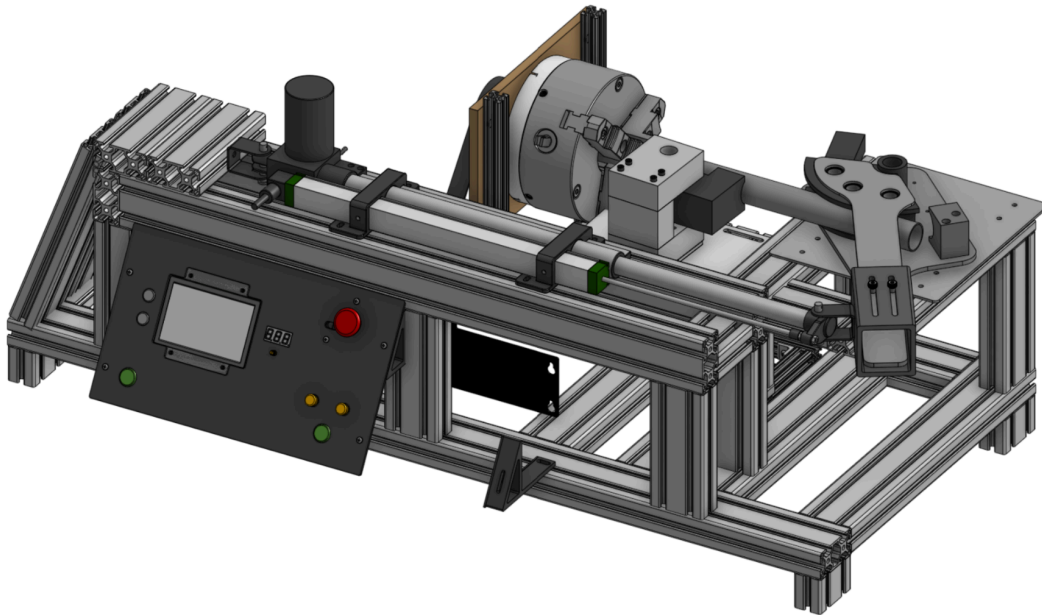
TABLE OF CONTENTS

Mechanical Design	2
Assembly Overview	2
8020 Framing	3
User Interface	4
Force Calculations	6
Bending	8
Clamp Die	8
Bending Die	9
Moment Arm	10
Advancing	11
Gripping	12
Axial-Rotating	14
Critical Hardware Selection	15
Electrical Design	16
Design Overview	16
Power	17
Actuators	18
Safety Components	25
User Interface	30
Circuit Board	33
Software Design	39
General Software Design	39
Mux Class:	39
REncoder Class:	39
Explanation of Overall Software Decisions	40
Communication	40
Config Class:	40
Testing/Version Control:	40
Next Steps	41
Appendix	42
Appendix A1. MATLAB Code	42

Mechanical Design

Assembly Overview

The Top Level Assembly (TLA) of Tubender consisted of three mechanisms that were further divided among the mechanical engineers. The TLA is pictured below:

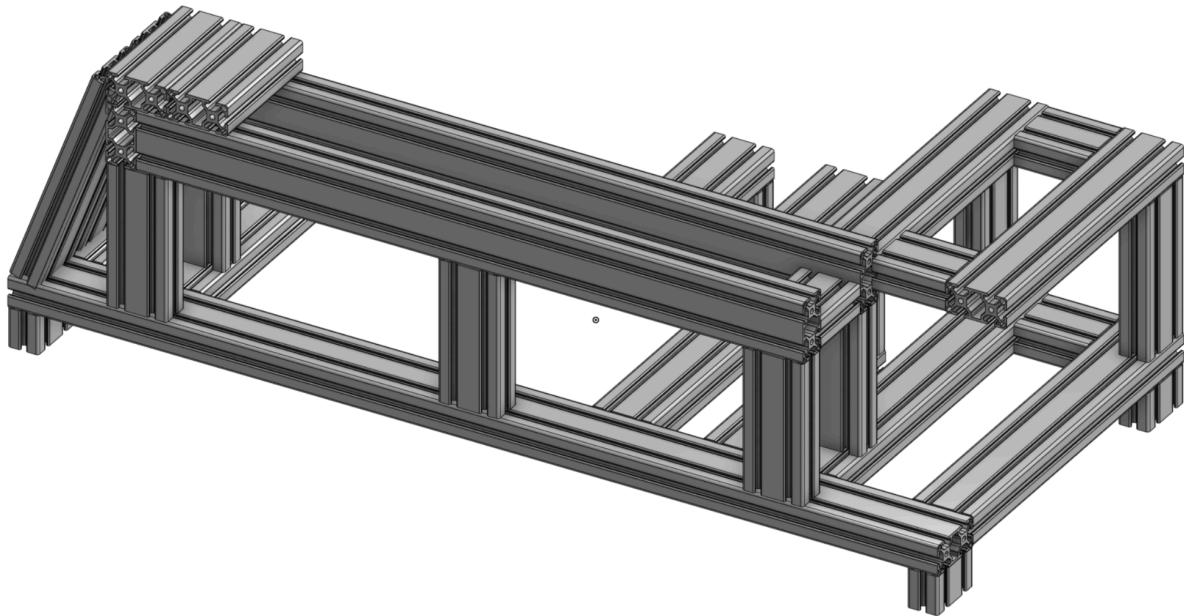


These mechanisms included Advancing, Rotation/Gripping, and Bending.

The design excels with its capability to perform three-axis bending using a three-point bending approach, specifically tailored for 1.5-inch diameter tubing, while maintaining robust and efficient features for precision and ease of use. In the bending process, the moment arm maximizes force from the linear actuator, while the clamp die ensures a second point of contact to prevent kinking, and the pressure die adds a third point of contact. The base and connecting plate are designed for quick and easy swapping, improving modularity as much as a first prototype can. For advancing, the platform effectively connects the lead screw to the framing and the rotation mechanism, enabling smooth motion. The grip and rotation mechanism is actuated within a single subassembly, streamlining functionality. The small form factor ensures the design aligns with the client's business objectives, providing a compact and efficient solution. However, the design does come with some limitations: it is restricted to bending only 1.5" diameter tubing, and easy modularity falls outside the current scope. The dimensions of the

machine are 48 inches long by 20 inches high by 22 inches wide, constrained by the framing and the height of the chuck.

8020 Framing



The 8020 Framing was the bulk of the support required for Tubender's subsystems. Due to the the t-slotted rails, adjusting the location and placement of various mechanisms was time-consuming but ultimately allowed for variability and tolerance issues to be mitigated. Though we chose the 4080 profile for its strength, it was difficult to work with due to hardware shortages, placement with the double profile, and more expensive. Additionally, the strength trade off did not solve the issue of force distribution, and was unable to sustain the reaction force load from the moment arm and linear actuator. The next iteration should use a stronger material for the framing in high impact areas, and perform the necessary yielding calculations (i.e. square steel tubing). The trade-offs of this approach is it's more expensive and mounting/connections would be more difficult.

Other areas of improvement would be starting with the framing and designing the subsystems around it rather than the other way around, as well as importing the fastening brackets and bolts into the assembly (at least in key areas) to determine if any interference would occur during the assembly process.

User Interface

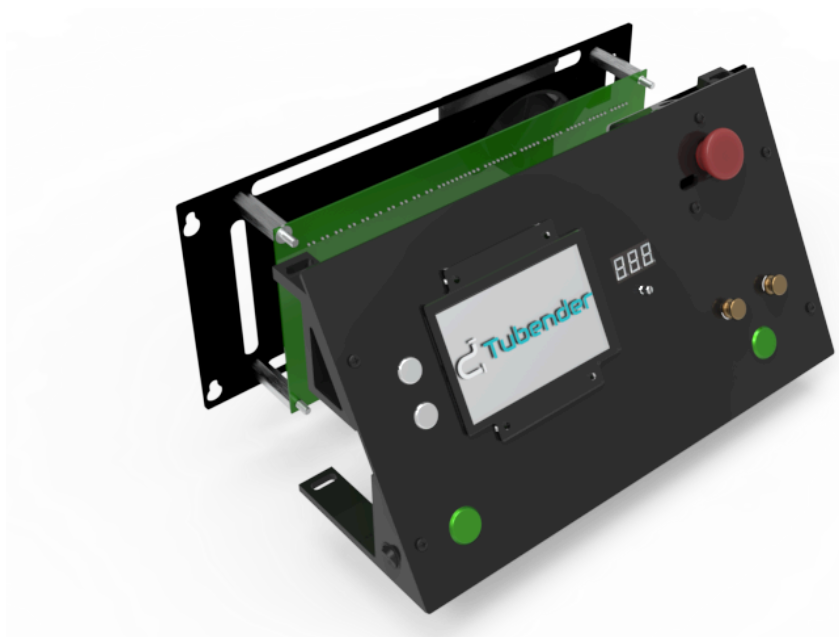


Figure _: Keyshot Rendering

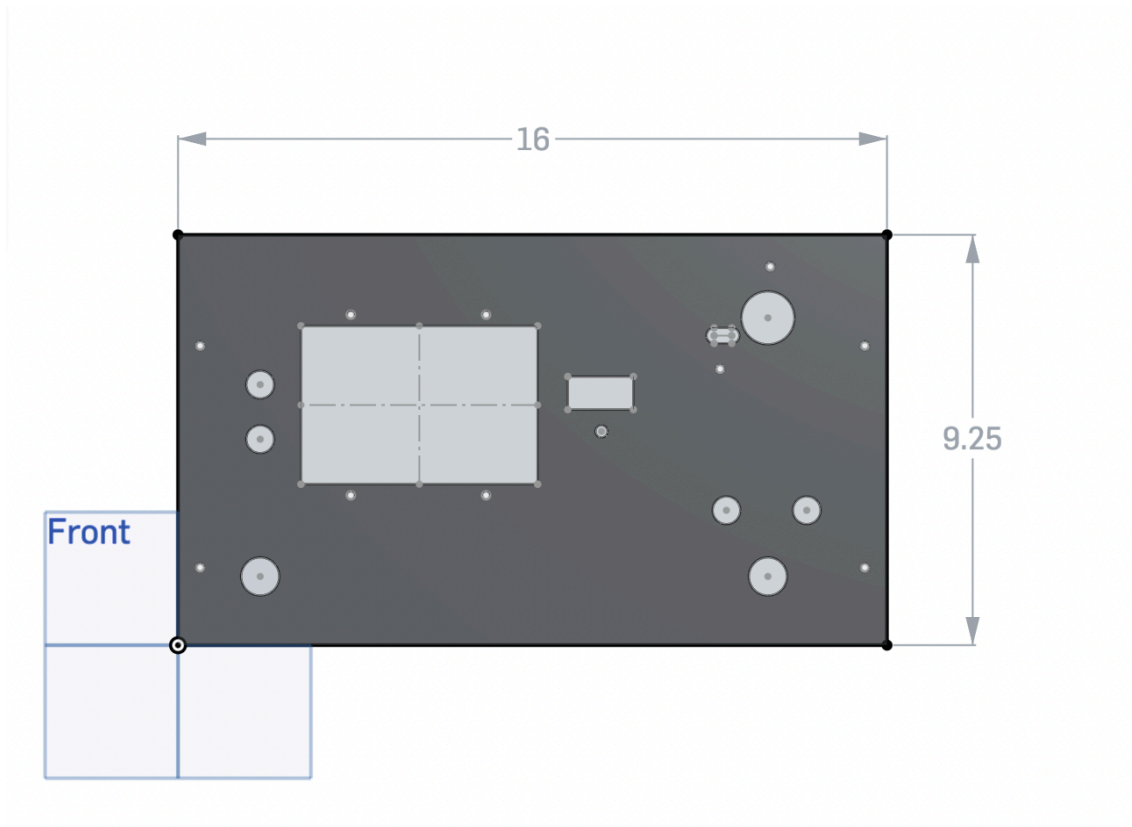
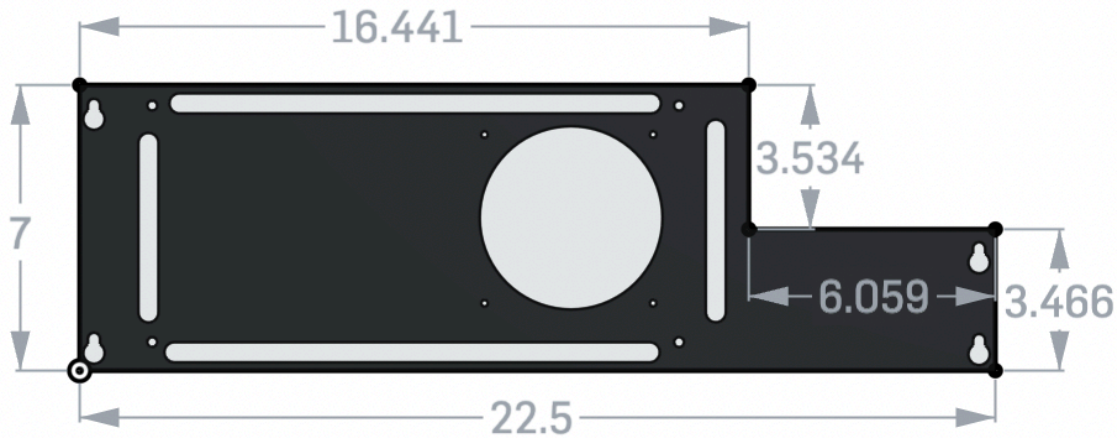


Figure _: Front Plate



Figure_: Back Plate

The front plate was designed to mount and display the buttons, UI screen, 7 segment display, and emergency stop in one, localized area. An $\frac{1}{8}$ " acrylic board was laser cut and mounted, on a 60 degree angle, to the 8020 framing. As this component is the main system the user interacts with, we decided that visual aesthetics is important. Due to this, we chose black matte acrylic for the front plate.

The back plate was designed to house the main PCB. It was connected to the PCB via 1" standoffs to ensure accessibility. Then, the backplate was mounted to the 8020 framing via T-slots. Because we anticipated maintenance to be conducted on the PCB, we introduced T-slots as a way to ensure easy removal. The large circle on the backplate is for a 120mm fan in order to cool down the PCB. Because the backplate is somewhat hidden within the system, we laser cut $\frac{1}{8}$ " MDF and spray painted it black, with the highest consideration being cost.

In the future, I would be interested in exploring different options for UI. A major limitation for this subsystem was time, and I would have liked to have more time to brainstorm and design multiple iterations. I would have liked to design one, a cohesive box that houses all of the electrical components, so the wires are not visible to the user. This is not only visually pleasing, but also safer to the user. This could have been achieved through 3D printing, though the size of the 3D printer was a limitation for us, or laser cutting and assembling a box. Though laser cutting a front and backplate was the most time and cost efficient method, I would be interested in exploring other options that would improve safety and visual aesthetics.



Force Calculations

Several force calculations were made to determine what was necessary to bend the tubing. These were reliant on several constants, like material properties and tube dimensions, as well as location from which the force was acting on (“moment arm”). We initially were set on bending steel, however due to time and price constraints optimized around aluminum, as the material is weaker (lower Young’s modulus) and therefore easier to bend.

The force calculations determined the proper torque specifications for the motors, with the highest force being required for bending the tube. Initially, the design relied solely on the rotation of the bending die to achieve the necessary torque. However, this approach required a 1 hp motor, which was impractical due to its high cost and the need to gear it down significantly to match the required RPMs. No commercially available gearbox could achieve the required reduction ratio. This limitation led to the concept of incorporating a moment arm, which effectively breaks the torque requirement into two components: force and distance. A heavy-duty linear actuator provides the necessary force, while a physical steel arm attached to the bending die serves as the moment arm to increase the effective torque. Since torque is defined as the product of force and distance, increasing either the applied force or the length of the moment arm proportionally increases the available torque for bending. This configuration reduces motor load, enabling the use of a more viable linear actuator while meeting the required torque for tube bending. The calculations can be seen as follows:

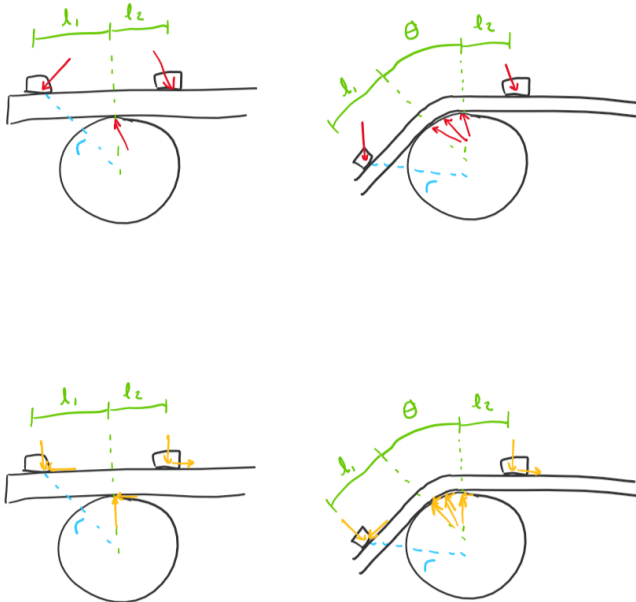
If just point force

$$M_{max} = F \cdot \frac{l_1 \cdot l_2}{l_1 + l_2}$$

$$R_{1,max} = F \cdot \frac{l_2}{l_1 + l_2}$$

$$R_{2,max} = F \cdot \frac{l_1}{l_1 + l_2}$$

→ MBLAD: 12,010 in-lb ~ 720 W @ 5 rpm
 l_1 and l_2
 don't matter or 0.952 hp



The relationship between the linear actuator stroke length (Y-axis) and the bending angle (X-axis), based on CAD dimensions, indicates that the maximum theoretical bending angle is approximately 56 degrees. However, to account for safety and practical limitations, it is more appropriate to assume a stroke length closer to 14 inches rather than the full 16 inches, which corresponds to a bending angle of approximately 51 degrees.

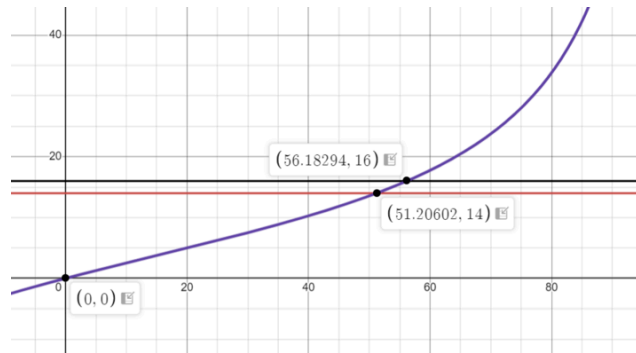
For testing purposes, it is recommended not to exceed a bending angle of 45 degrees, as this likely aligns with a standard bend angle. The reduction from the originally intended 90-degree bending angle is due to the necessity of moving the linear actuator farther from the center of the die (distance L) and decreasing the initial angle of the bend arm (θ_0). These adjustments were required to avoid interference with the die clamp, which was not initially accounted for in the preliminary dimension calculations. New calculations are shown below:

$$x(\theta_{tot}) = L \left[\tan \theta_0 + \tan(\theta_{tot} - \theta_0) \right]$$

From CAD:

$$L = 14''$$

$$\theta_0 = 15^\circ$$



Bending

Clamp Die

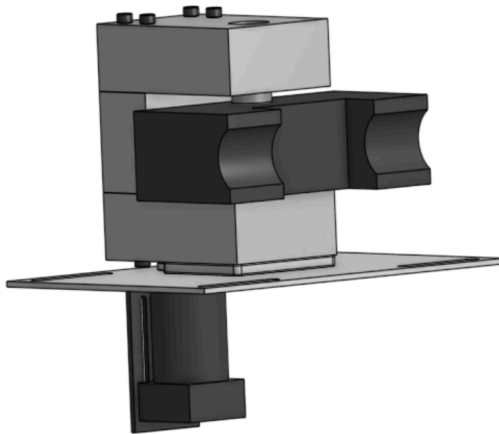


Figure X: Clamp Die Assembly

insert CAD

The clamp die was a component in the Tubender system, designed to secure the tube during the bending process. Its primary purpose was to prevent the tube from rotating, and moving forward, or deforming under reaction forces, while also guiding the tube through the system and preventing kinking. To achieve this, a 153x78x50.5mm steel clamp die, specifically designed for a 1.5-inch outer diameter (OD) tube, was integrated into the assembly.

The clamp die was mounted onto a fabricated aluminum "clamp arm" using a dowel pin, which allowed it to rotate as required. This clamp arm connected the die to a motor, enabling it to clamp the tube securely when positioned for bending and to unclamp when additional tube extrusion was needed for subsequent bends. The assembly was fixed to the framing structure using a laser-cut acrylic plate, which provided a secure mounting base. To ensure smooth rotation of the aluminum arm, a 3D-printed resin block was fixed between the acrylic plate and the arm, effectively reducing friction during rotation.

During operation, the motor rotated the clamp arm, positioning the clamp die to hold the tube firmly in place. As the tube extruded forward, the die, mounted via the dowel pin, could rotate out of the way, allowing the extrusion to continue without obstruction. However, one limitation of this design was that the aluminum arm exerted pressure on the framing during the bending process, causing the framing to bend. This revealed a structural weakness in the framing material. To address this issue, future iterations should incorporate stronger framing materials to enhance durability and maintain the structural integrity of the system during operation.

Bending Die

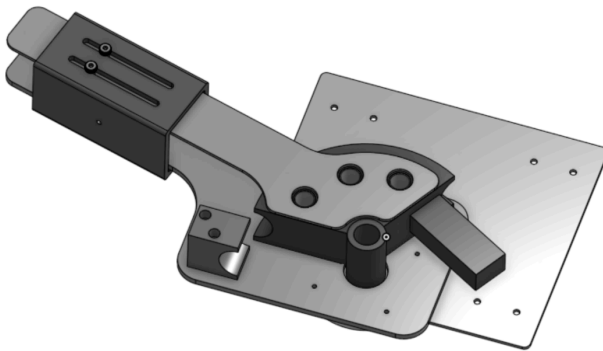


Figure X: Bending Die Assembly

The bending die subsystem was designed because it was the most effective solution for bending the tube within the constraints of the project's budget and timeline. This subsystem included several key components: a bend die designed for 1.5-inch outer diameter (OD) tubing, a pressure die fabricated from steel for the same tubing dimensions, and a baseplate and connecting plate joined by a lazy susan bearing.

The bend die had four holes, one serving as the pivot point for rotation and the remaining three used to secure the die to the assembly through the connecting plate with press-fit pins. The pressure die, mounted to the connecting plate using screws, applied consistent pressure to the outside of the tube while the connecting plate rotated. This pressure created the necessary force to achieve the desired bend shape while the tube was rotated by applying constant pressure which also prevented wrinkles, and uneven bending along the tube. To attach this subsystem to the assembly, a lazy susan bearing connected the connecting plate to a baseplate, which was fixed to the system's framing.

Moment Arm

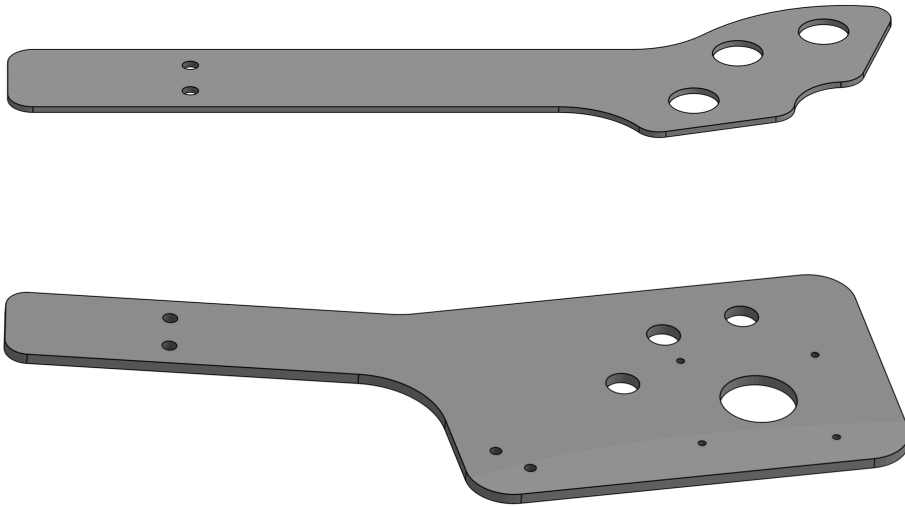
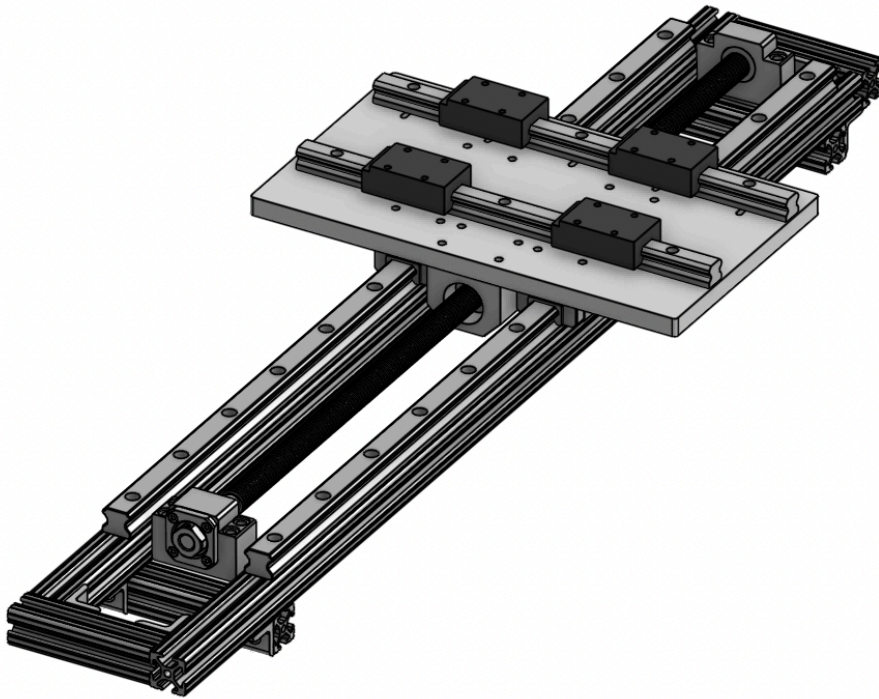


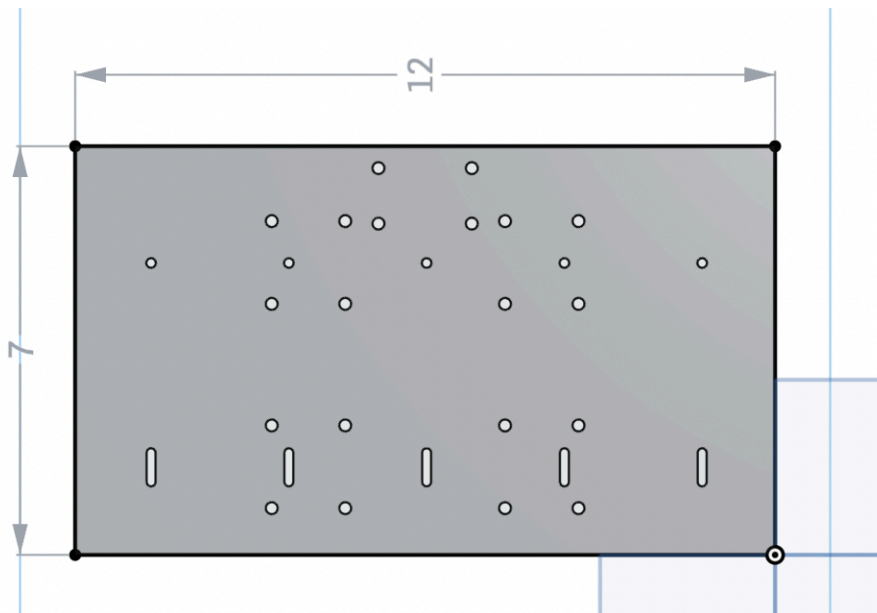
Figure X: Moment Arm

Figure X: Rotating plate

The main piece of the bending subsystem was the moment arm. It was set at a 15° angle and was 12" long due to positional constraints from the clamping die. In the final assembly there were two different arms that made up the system, a standalone arm and one that was integrated into the rotating plate. Around the moment arms was what we called a "bracelet." This was a 3"x3"x6" box steel piece that we created slots for shoulder screws and a tapped hole to attach it to the linear actuator. The slots are what allowed the linear actuator movement to be used as rotational movement. The moment arm itself allowed us to reduce the forces needed to bend the tube.



Figure_:Advancing Assembly CAD

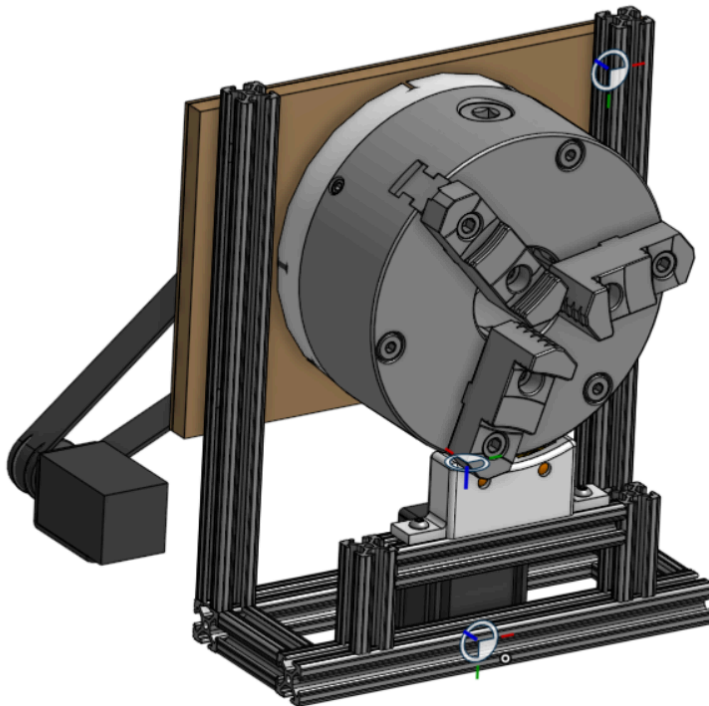


Figure_: Platform Overall Dimensions

The advancing sub-system was designed to advance and retreat the rotation assembly. The platform sits on top of 4 sliders that advance along the longer rails shown in the photo of the advancing assembly. On top of the platform, there is another set of rails and 4 sliders, perpendicular to the advancing rails, where the rotation assembly is mounted. The perpendicular sliders & rails account for, and try to absorb, some of the force generated when bending a tube.

In the future, I would have created bend diagrams and done more calculations to be more certain about the requirements and design of the advancing system. For the platform, we assumed $\frac{1}{4}$ " aluminum would be able to sustain the weight of the chuck, and it did, but further bending moment diagrams would not only make me feel more comfortable, but also optimize the thickness of the platform. In other words, by doing bending calculations, I could have found a smaller thickness than $\frac{1}{4}$ ", thus reducing costs. In terms of the perpendicular rails, I decided on the length of the rails based on videos and demonstrations of other autonomous bending machines. In the future I would like to calculate the actual force and displacement a bending tube would create at the location of the gripper and incorporate this into the design of the advancing system.

Gripping



For the rotational subsystem, the main component that was built around was an 8 inch lathe chuck with a through hole to accommodate the 1.75" Outer diameter EMT Conduit tubing.

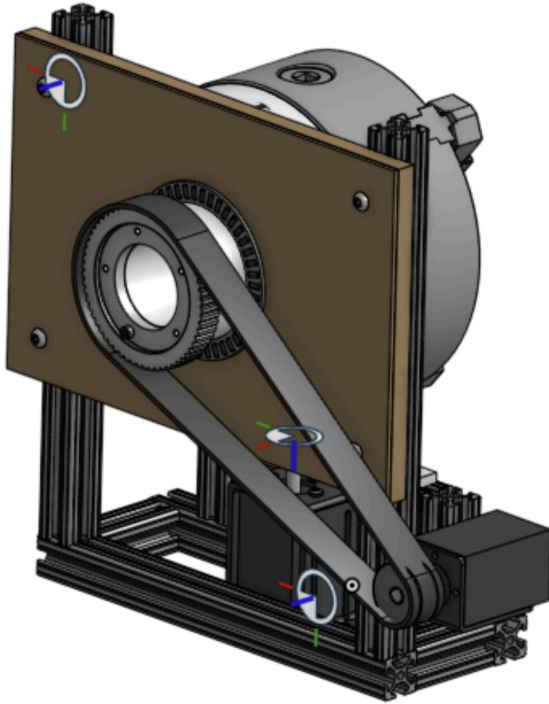


This chuck was mounted on a custom support, which was 3D printed in ASA with 60% in fill. The framing was mainly 8020, with a front flange seen in grey to hold some of the load of the chuck, which was approximately 40 pounds. This front flange was made of ASA, with two steel dowel pins to hold two small bearings in place. These bearings facilitated rotation in combination with a larger bearing press fit into a wooden back plate. From Front to Back, the stack up is as follows: 8 in Lathe Chuck, custom chuck support, thrust bearing, smaller retaining wall made of $\frac{1}{8}$ " MDF, press fit main bearing in a custom $\frac{1}{4}$ " back plate, and a final thrust bearing.

Below the chuck on the center line, you can see a 100:1 gear ratio stepper motor. The goal for this was to create a custom, automated chuck clamping system. We attempted to code a system where a custom female chuck key insert would be pressed into the traditional chuck key slot. The chuck would "home" to the 0 position unclamped, where the key would interface with a male chuck key extension off of the shaft of the stepper motor. Both of these keys were machined from high carbon steel. In theory, this idea worked wonderfully, but in practice the homing and calibration would be time consuming, and became a low priority when time became a constraint in our design. If I had infinite time and extensive sensor knowledge, this idea would have absolutely been doable in terms of creating an automated system for clamping. However, a lack of expertise, budget, and time made this homing and clamping system prove more difficult than previously intended.

To make this project with no budget constraints, tighter tolerancing and increased strength of materials would be a necessity. Hand cutting 8020 supports for our system proved difficult to make items fit perfectly. The tolerance on the custom chuck support could have been better, as the bearing retention proved to be difficult. Overall, all of our custom pieces could have been machined from higher strength steel to help support some of the larger stresses that would have come from bending such large tubes.

Axial-Rotating



The purpose of the axial rotation mechanism is to rotate the tube intermittently and allow the bending to be done in different x-y-z planes. The client requested that we bend the tube in my multiple locations to allow for full automation of the entire tube rotation and not need to remove and reinsert the tube in the machine to get other angles bent.

This design contains 2 pulleys with the bigger one being 2 inches wide to fit the tube through it,, a timing belt, a motor (specified in the electrical section below), bearings, and the chuck back plate. We started designing the pulley + belt subsystem to use a motor rotation to rotate the entire chuck which is holding the tube. The chuck is mounted on a support that allows it to rotate freely with the help of two bearings under the chuck. We picked a pulley and belt system over the sprocket and chain because it was within our budget and still strong enough for our rotating purposes, and the aluminum pulley we found was easily machinable using the machining resources we had on campus. We designed the chuck back plate such that we were able to screw the bigger pulley to it and had it sticking out of the back rectangle plate that is holding the chuck horizontally. Inside the back horizontal plate hole, we added a bearing to allow the chuck back plate to be flushed in and rotate. We also added a thrust bearing between the chuck back plate and the horizontal plate to facilitate the rotation of the chuck with less friction. The smaller pulley is attached to the motor's shaft on the side of the chuck 8020 support.



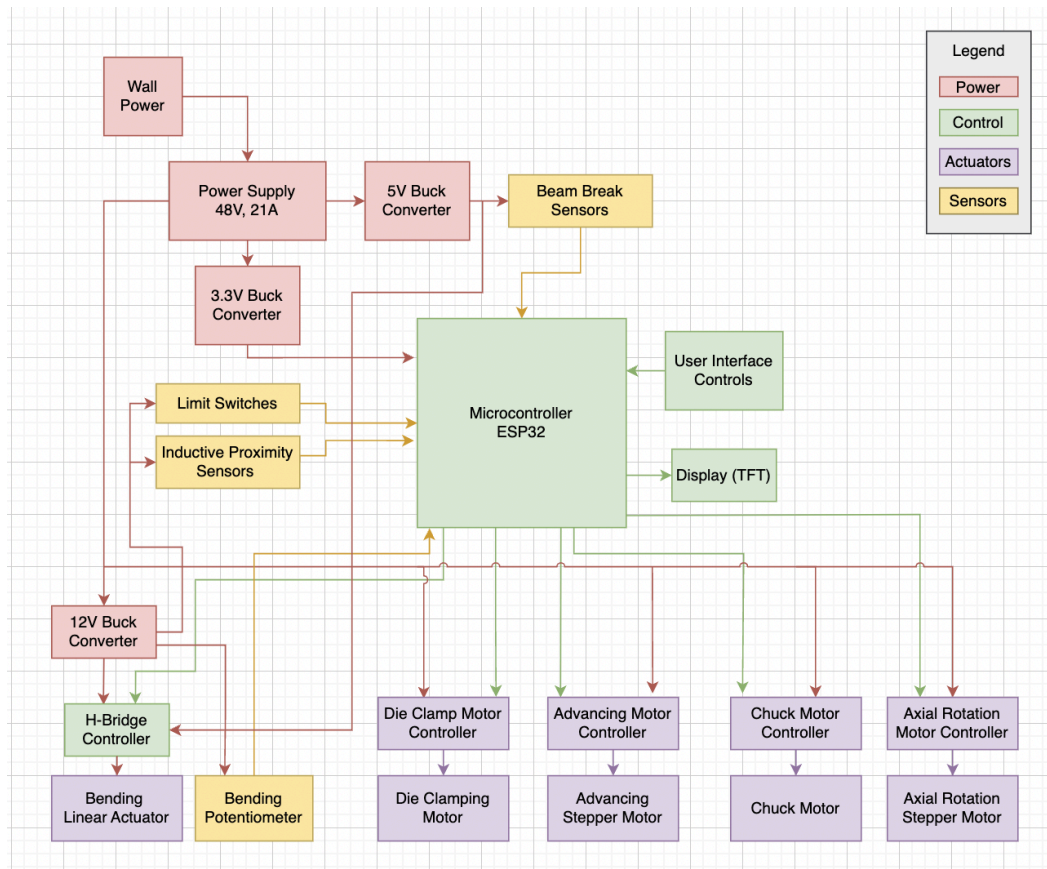
For future design improvements, I would recommend designing a stronger motor mount to the side, the one we 3D printed broke for being too thin and weak. I would also get a belt with a larger diameter and attach the motor to the platform and not the side of the chuck support- this will give it more stability and durability.

Critical Hardware Selection

Include the names and part numbers of the hardware involved in the assembly. This can be in the form of a table

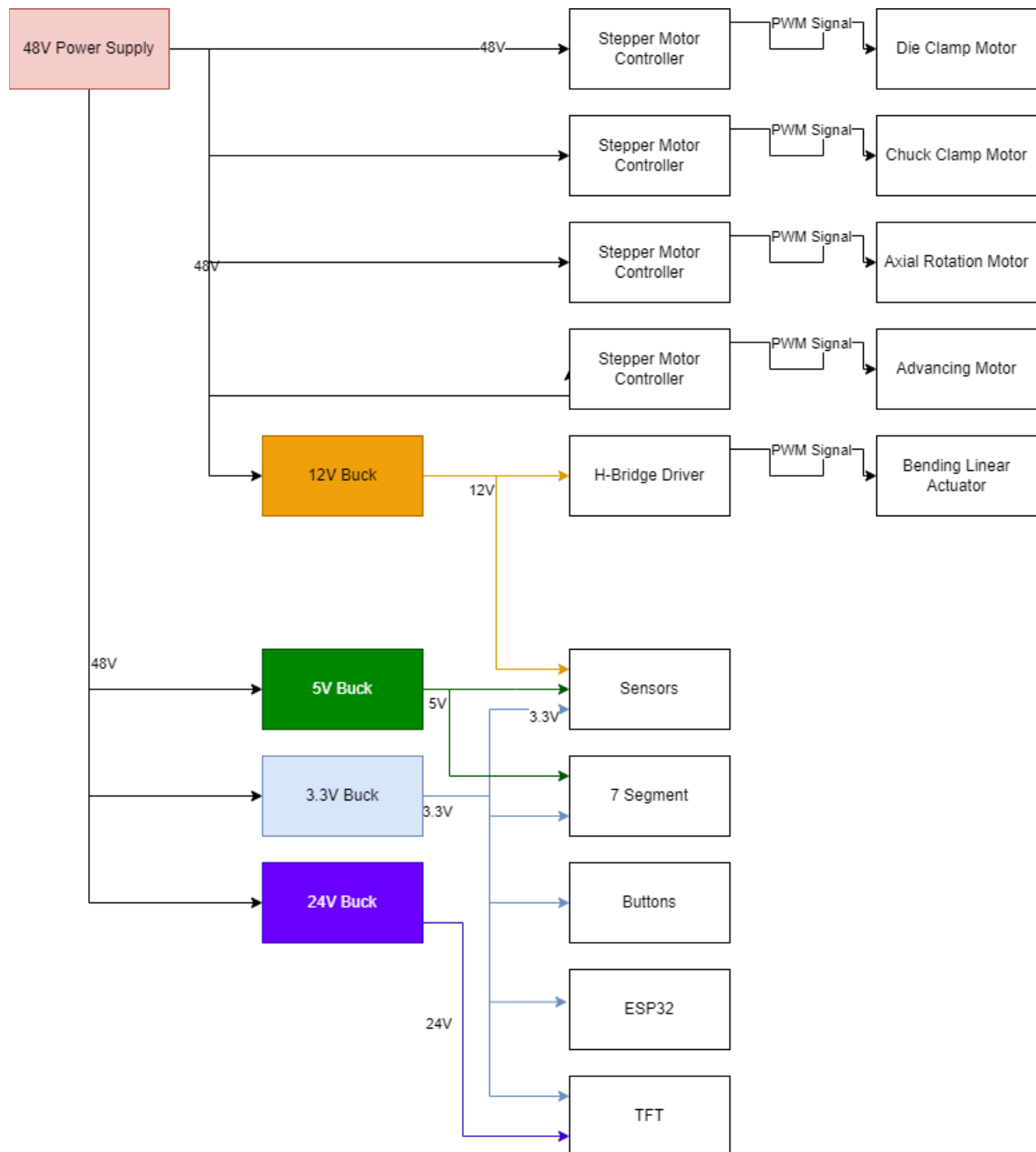
Electrical Design

Design Overview



The Tubender electrical system was designed to enable the different functions which outline the nature of the product. In terms of tube handling, there was a need to advance, rotate and clamp the tube, translating to four motors and subsequent drivers as well as, for the bending itself, a linear actuator and its own personal driver. In terms of safety, beam breaks, limit switches and inductive proximity sensors were chosen to ensure safe usage of the machine. In addition, to run and power everything a ESP32-WROOM-32UE-16 was employed as the MCU with 48V to 3.3V, 5V, 12V, and 24V circuits for bucking down to the necessary voltage rails. For user interaction, a TFT screen as well as a 7 segment display, rotary encoder and a suite of buttons were spec'd out and integrated into the machine.

Power



The Tubender power system is made up of a 48V power supply connected to wall power at 120VAC, which is further connected to four buck converters at 3.3V, 5V, 12V, and 24V. The buck converter topology was used over a linear regulator topology due to high voltage level conversion efficiency.

Purpose

The subsystems of Tubender require different voltage levels for power. The buck converters implemented are required to change 48V from the power supply to the voltage levels different parts require. The 3.3V buck converter provides power to logic level components, including some sensors, the ESP32 microcontroller and GPIO expanders, the TFT display, the UI buttons and rotary encoder. The 5V buck converter provides power for some sensors and the seven segment UI display. The 12V buck converter provides the voltage for the bending linear actuator, and the 24V converter provides additional power to the TFT display circuitry.

Placement

The power network is placed in close proximity to the PCB power supply input. This minimizes thermal or power losses on the highest voltage level net.

Actuators

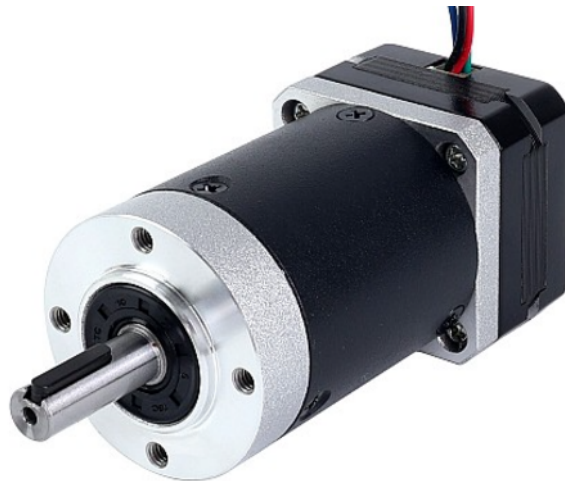


Figure _: Nema 17 100:1 Stepper Motor

Purpose

The 100:1 stepper motor is used in three forms of movement within the Tubender machine. The first motor is used for axial rotation of the tube, another is for the clamping of the chuck, and the last is for clamping of the bending die. We decided on a stepper motor for all three movements due to its combination of accuracy, simplicity, and cost-effectiveness when compared to a servo motor. The Nema 17 stepper motor with a 100:1 planetary gearbox allows for high torque and positional accuracy, which fit the cases of axial rotation and clamping to ensure they would be

both powerful and accurate enough to overcome the torque for each of those moments, while still maintaining accurate stopping points, like rotating to an exact 90 degrees. This motor was bought from Stepperonline, with solid documentation for the motor specifications and dimensions.

Placement

The first 100:1 motor used for axial rotation is placed on the side and below the chuck, placed within a pulley system to drive the rotation of the tube. The chuck clamp motor was placed directly below the chuck, and the die clamp motor was placed beneath the die clamp, connected through a clamp arm to allow for secure clamping.

Spec Sheet and Notes

All applicable information for this motor is found on the Stepperonline page linked [here](#). This motor runs off of 24V DC power, with a rated torque of 15 Nm, which is sufficient for all three movements we used this motor for.



Figure _: Nema 17 10:1 Stepper Motor

Purpose

This motor is used to power the advancing movement of the Tubender machine. Similar to the 100:1 motors, we chose a stepper motor for this movement due to its efficiency and accuracy, and since this movement doesn't require a large amount of torque, this motor is a great choice for our purposes. It has increased torque and high precision, which was just enough for the small load of the advancing and the accuracy needed for this movement. This motor was bought from Stepperonline, with solid documentation for the motor specifications and dimensions.

Placement

This motor sits at the end of the machine's frame, placed right behind the lead screw to power and advance the chuck rotation and grip subsystem. A shaft collar of appropriate sizing was attached to the motor to then integrate into the lead screw system, allowing for advancement of the system.

Spec Sheet and Notes

All applicable information for this motor is found on the Stepperonline page linked [here](#). This motor runs off of 24V DC power, with a rated torque of 5 Nm, which is sufficient for the advancing movement we used this motor for.

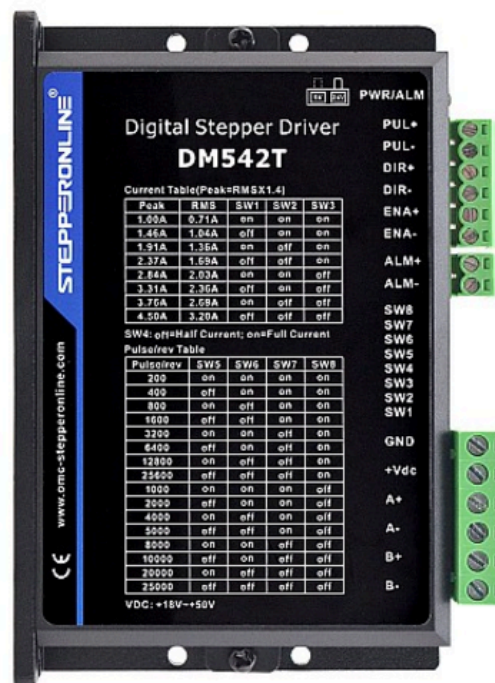


Figure _: DM542T Digital Stepper Driver

Purpose

The motor controller is the hardware used to interface with the four stepper motors to control when to move, speed, and direction of movement for these motors. We have one motor controller used to control each stepper motor, for a total of four motor controllers.

Placement and Connections

All 4 motor controllers were placed next to the PCB, hidden in a corner of our 80/20 framing, with the wires running through a triangle of 80/20 to stay hidden and close to the frame, ensuring no collisions or contact would be made with these wires and the function of the machine. The wires were threaded through a mounting hole created specifically for the wires, and then the connector at the end of the motor wires was plugged into the PCB. For the connections, the +Vdc terminal should be connected to 48 V with 18-22 AWG wire. All ground terminals including PUL-, DIR-, ENA-, and GND are connected through a Wago wire connector, and then to ground. PUL+, DIR+, and ENA+ were all crimped and placed into a 5 terminal connector, with the other 2 terminals being filled with the GND wire and +Vdc wire. Lastly, the A+, A-, B+, and B- motor signal wires are connected straight to the driver using 22 AWG wire, making sure the wire was long enough to reach the drivers. Zip ties and labels are used to organize the wires and make connections easier and quicker for situations when disconnections need to be made.

Spec Sheet and Notes

The information and data sheet for the motor drivers can be found [here](#). The data sheet is very informative and explains the purposes of each pin, as well as how to set up the switch configurations to set output current, idle current, and microstep of each motor. We set up all four motors to run at their corresponding peak currents and at 200 steps/revolution.



Figure _: Bending Actuator

Purpose

The bending actuator provides the linear force that pushes the bending arm to rotate the bending die - bending the tube. It was calculated that the required force would be roughly 1000 lbs, so this linear actuator was chosen because of its 1300 lbs rating. A linear actuator was chosen over a rotational actuation method for bending because of the incredible torque requirement to bend a steel tube. We were unable to find gearbox options that were rated for this torque within our budget or with a lead time that would be reasonable for one semester. This would be one area of future improvement, though, as a rotational mechanism would simplify the design of several components on the machine.

Placement and Connections

The linear actuator is connected directly to the linear potentiometer using the following 3D printed brackets. Two of these brackets connect the body of the linear actuator and linear potentiometer together and fasten them down to the 8020 framing beneath them. The third part connects the rods of the linear actuator and linear potentiometer so that the linear potentiometer can read the position of the linear actuator for precise movement using PID control. The ends of the linear actuator were connected to C-connectors with pins to hold them in place. The back C-connector was attached to the 8020 frame and the front C-connector was attached to the bending arm to rotate the bending die when extended. The linear actuator has two wires: power and ground. These polarities are reversible; when 12V is supplied to one of the wires and ground to the other, then it will extend. Reverse these connections, and the linear actuator will retract. These wires are connected to the BTS7960 motor controller.

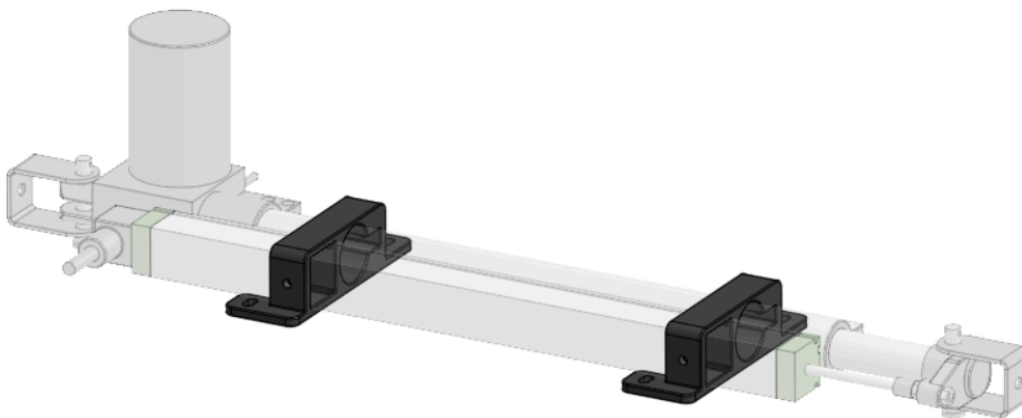


Figure _: Bending Actuator Mounting

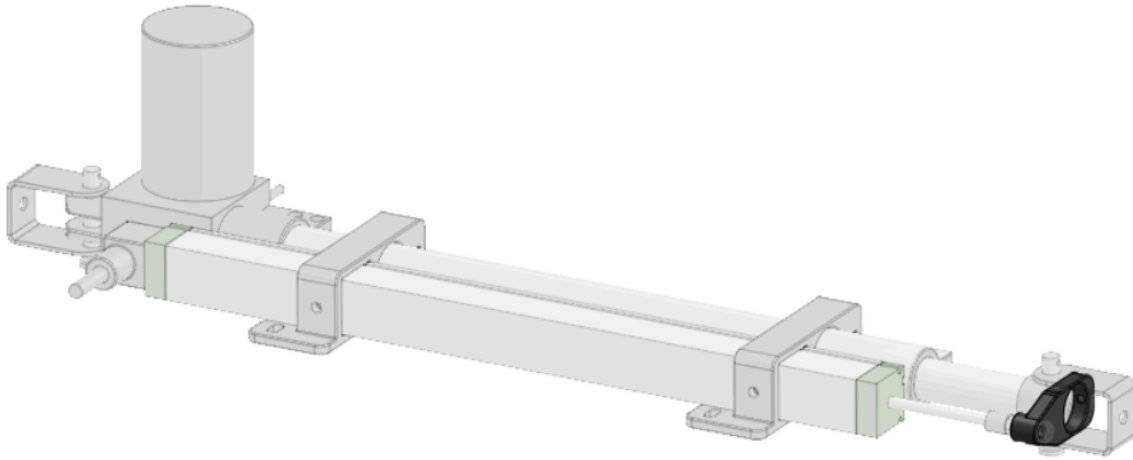


Figure _: Bending Actuator Rod Connector

Spec Sheet and Notes

All applicable information for this linear actuator is found on the Vevor page linked [here](#).



Figure _: Linear Potentiometer

Purpose

The linear potentiometer measures the position of the linear actuator in order to give precise feedback for the PID controller that controls the linear actuator. Precision was critical for the bending subsystem so that the machine could match the bend angle input by the user with a minimal margin of error.

Placement and Connections

The linear potentiometer was connected to the linear actuator using the 3D printed brackets seen in the above figures. Two of these brackets connect the body of the linear actuator and linear potentiometer together and fasten them down to the 8020 framing beneath them. The third part connects the rods of the linear actuator and linear potentiometer so that the linear potentiometer can read the position of the linear actuator for precise movement using PID control. We wired it slightly differently than a typical potentiometer that would have 3 wires: power, ground, and wiper. We only connected the wiper and ground pins to use it as a variable resistor - which worked as we initially intended. This was done because the extreme limit of the linear potentiometer had very low resistance (~15 Ohms), and in the original 3 wire configuration this would draw incredibly high currents. Instead, we used the variable resistance of this potentiometer as the second resistor in a standard voltage divider circuit, which resulted in a far more manageable current draw.

Spec Sheet and Notes

The website listing can be seen [here](#), which contains specification information.

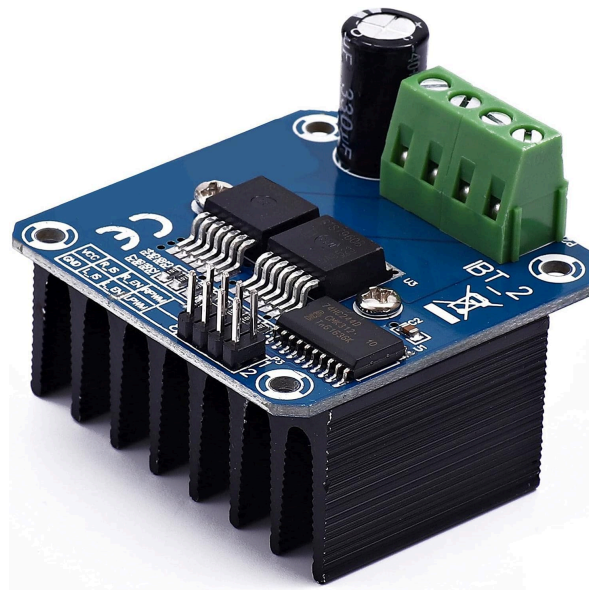


Figure _: BTS7960 Motor Controller

Purpose

The BTS7960 motor controller is an H-bridge controller that was used to drive the linear actuator. This type of motor controller was chosen because of its compatibility with DC motors, which is the motor type within the linear actuator, and its high current rating of 43A - well over our expected draw of 8A.

Placement and Connections

Mounting this component was a lower priority as Showcase neared, so it ultimately sat on the table below the PCB. There are 10 relevant pin connections on this board. The two linear actuator wires are connected to M+ and M- in either configuration since they're reversible. From the PCB, 12V goes to B+ and ground goes to B-. On the front of the controller, Vcc is connected to 5V and gnd to ground. The remaining pins, L_EN, R_EN, L_PWM, and R_PWM are all connected to GPIO pins on the PCB.

Spec Sheet and Notes

More information about the motor controller can be found [here](#). Although the documentation on this page may not be incredibly robust, there are also many helpful YouTube videos that explain how to use this motor controller, such as [this](#).

Safety Components



Figure _: Unclamp Button

Two large high visibility buttons for interrupting operation to release machine clamps were installed to prevent the user or unintended materials from being trapped in either clamp. Placing additional unclamp buttons next to the respective chuck or die clamp was an additional safety oriented design choice that was formulated but not implemented.

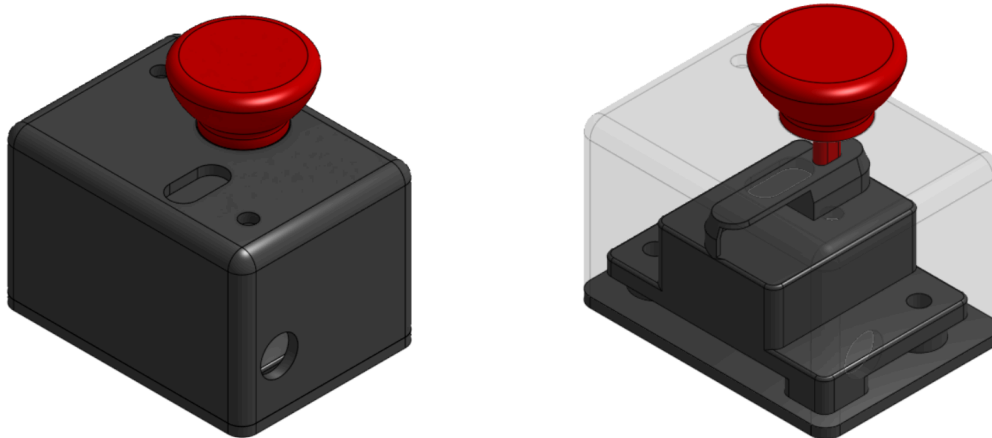


Figure _: E-Stop

Purpose

Emergency stops are a critical component for industrial equipment to ensure the safety of its operators. When the e-stop button is pressed, power is disconnected from the whole machine, turning everything off.

Placement and Connections

The e-stop was mounted to the UI panel at the front of the machine. The intention is to make this button as accessible and intuitive as possible, especially since it is most needed in dangerous scenarios when something has gone wrong. The e-stop is wired in series with the 48V line between the power supply and the PCB so that power is cut to the machine when the button is pressed.

Spec Sheet and Notes

For this project, we opted for a circuit breaker that is rated for 100A of current, which can be seen [here](#). Most standard e-stop buttons are only rated for 10A of current, which did not have a high enough factor of safety based on our expected draw of 8A when operating the bending linear actuator. Because of this, we needed to design an enclosure and button that will replicate the user experience of traditional e-stops, while also allowing access to reset the circuit breaker. This design can be seen above.



Figure _: Limit Switch

Purpose

These limit switches were used as end-of-travel detection for the advancing subsystem and for the die clamp. When the advancing platform comes in contact with one of the limit switches on either end of the linear rail, the advancing motor is immediately turned off. The advancing limit switches are not intended to be pressed under normal operation, and exist primarily as safety features to prevent overtravel. Similarly, when the die clamp runs into either of its limit switches, the die clamp motor immediately turns off. However, these limit switches are intended to be used under normal operation.

Placement and Connections

The advancing limit switches are mounted at either end of the linear rail so that that platform will run into them before crashing into any other parts. The die clamp limit switches are mounted adjacent to the die clamp. One of the limit switches is placed so that it is pressed when the clamp is closed, and the other is placed so that it is pressed when the die clamp opens by a specified angle (for example. 45 degrees). All of the limit switches have two wires. One of the wires is connected to 3.3V, and the other is connected to a GPIO pin in parallel with a pulldown resistor to ground.

Spec Sheet and Notes

The spec sheet can be found [here](#). These limit switches were chosen to be normally open (NO) so that the electrical connection is only made when the button is pressed, meaning it can be connected in-line, which was a simpler option than its normally closed counterpart.

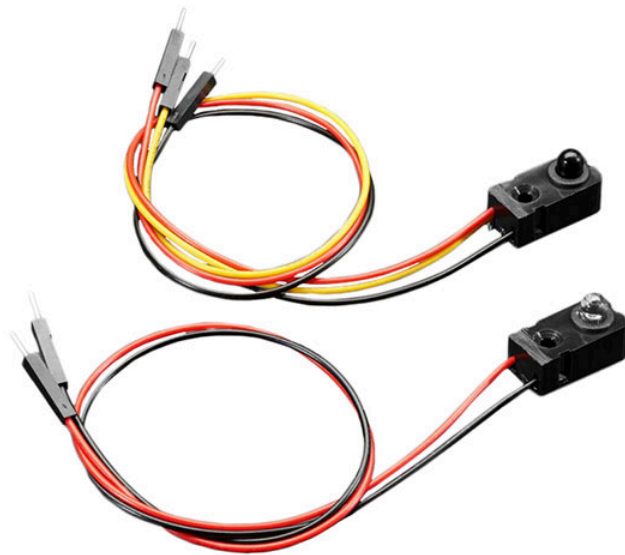


Figure _: Break Beam Sensor

Purpose

The break beam sensors were used to detect the presence of the tube in two different locations. If the tube is not detected, then certain actions in the code won't execute until the tube is properly inserted. For example, if the break beam behind the chuck isn't triggered, then the chuck motor will not rotate. This prevents the user from crushing their fingers if the chuck is tightening while a tube isn't present.

Placement and Connections

One pair of the break beam sensors is located behind the chuck to detect if a tube has been placed into the chuck. The chuck motor will not tighten if the break beam isn't triggered. Another break beam pair is placed right before the die clamp so the machine knows that there is a tube to clamp onto. For both pairs, the transmitter has two cables - red for 5V and black for ground. The receivers each have 3 cables - red for 5V, black for ground, and yellow for data that is connected to a GPIO pin.

Spec Sheet and Notes

More information about the break beam sensor can be found [here](#).

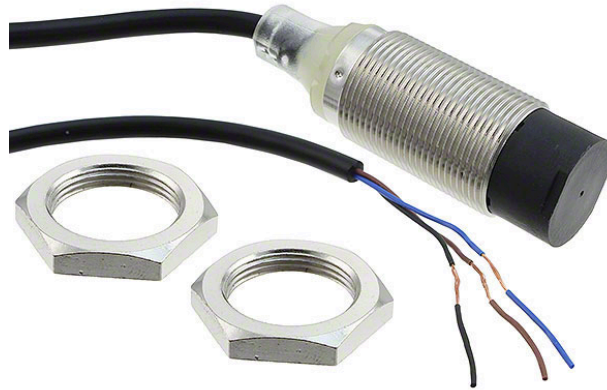


Figure _: Inductive Proximity Sensor

Purpose

The inductive proximity sensors are used to detect the presence of metal in front of them - in this case the tube. If the tube isn't detected on both sides of the chuck, then the chuck motor won't tighten to ensure user safety. If the tube is detected on both sides, then it guarantees the user doesn't have any fingers or other pinchable objects within the clamp's grasp, meaning it is safe to tighten.

Placement and Connections

The proximity sensors are placed on either side of the chuck at a minimum distance of 35mm to ensure no interference from the metal in the chuck. The tip of sensors are located 7mm or closer to where the tube would be located to ensure the aluminum in the tube is properly detected. Each sensor has 3 wires - brown for 12V, blue for ground, and black for data that was connected to a GPIO pin.

Spec Sheet and Notes

The datasheet for the inductive proximity sensor can be found [here](#). The unshielded option was chosen to maximize the range at which the sensor would detect the tube. This was especially important because the tube was made of aluminum, which is not ferromagnetic, causing detection to be weaker.

User Interface

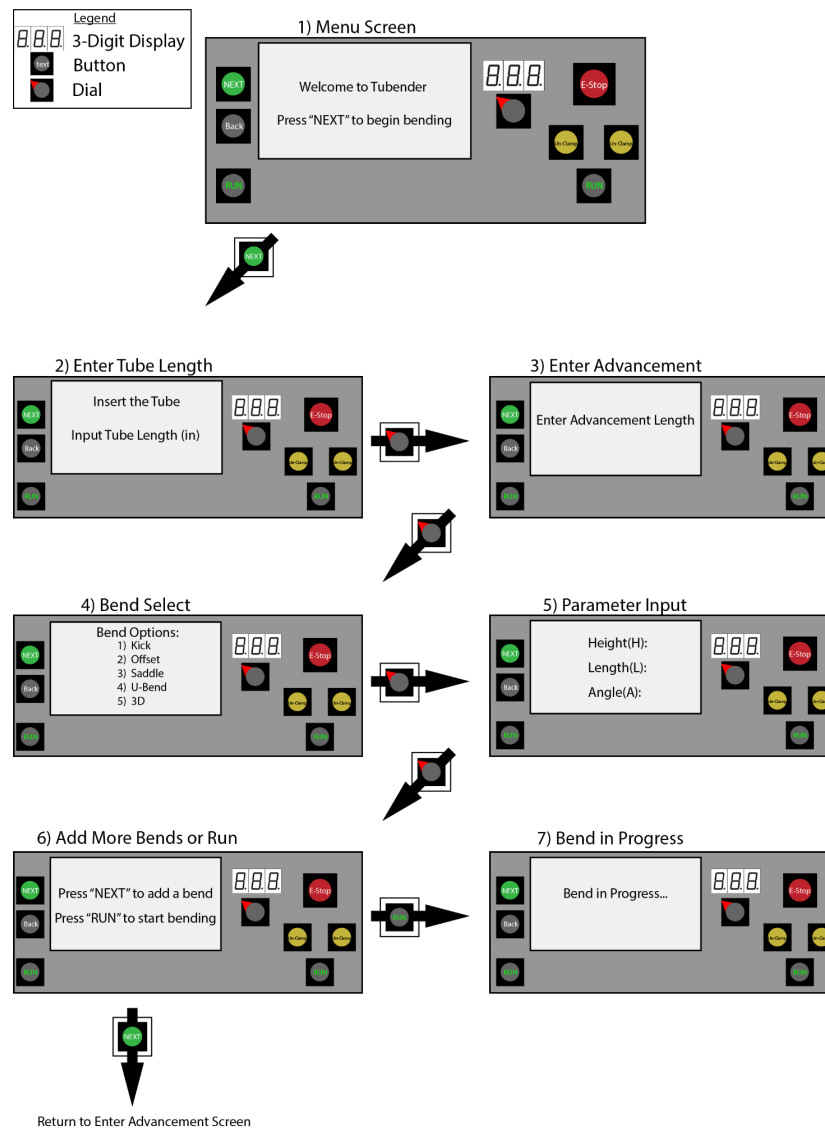


Figure _: User Interface Screen Map

The User Interface Screen Map depicts the flow of user inputs and information prompts. Each grey panel of the screen map represents a screen state with arrows in between specifying the required input for a transition between states. Most screens are designed to prompt the user for a numerical value, shown on the 7-Segment display and selected using dial rotation. Other screen navigation options include a next and back button for transitioning sequentially through bend input options and screens.



Figure _: UI Components (Left to Right: Unclamp Button, Power Switch, Next/Back Button, Run Button)

Purpose

Buttons of varying sizes and colors were selected to convey different functionality, such as the run buttons containing a green LED or unclamp buttons being visually distinct for emergency release. Additionally, all buttons receive hardware debouncing on the main board and are connected to a single 8:1 multiplexer to minimize GPIO usage.

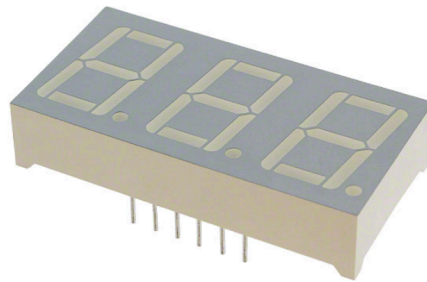


Figure _: Standalone 7-Segment Display

Purpose

A 7-Segment display was chosen in conjunction with a TFT display to provide a high refresh rate numerical display that allows the user to incrementally select a range of values for tube length, advancement distance, height, angle, and length. As regularly overwriting TFT pixel values is a time intensive process, a high speed shift register was selected for taking bytes of data from ESP32 Microcontroller, and writing a high or low value to each of the 7-Segment Anodes depending upon the desired digit value. In combination with a shift register, a 2 of 3 decoder was connected to the 7-Segment cathodes for rapidly selecting which of three digits is being written to, allowing positive numerical values of up to three digits to be entered.



Figure _: Rotary Encoder ([PEC11R-4215K-S0024](#))

Purpose

A rotary encoder was selected for incrementing and decrementing numerical values shown on the 7-Segment Display. In software, the A and B signal channels of the rotary encoder are compared to determine if the rotary encoder is stationary, rotating clockwise or counterclockwise.



Figure _: Adafruit 7-Segment Display with I2C Interface ([879](#))

Purpose

After facing software and hardware control issues with the shift register and decoder circuitry, a new 7-Segment display was selected for its simpler I2C interface. This display internally incorporates the shift register and decoder circuitry while allowing simple addressing and programming over I2C.

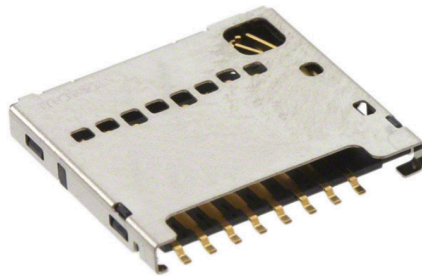


Figure _: Micro SD Card reader ([1040310811](#))

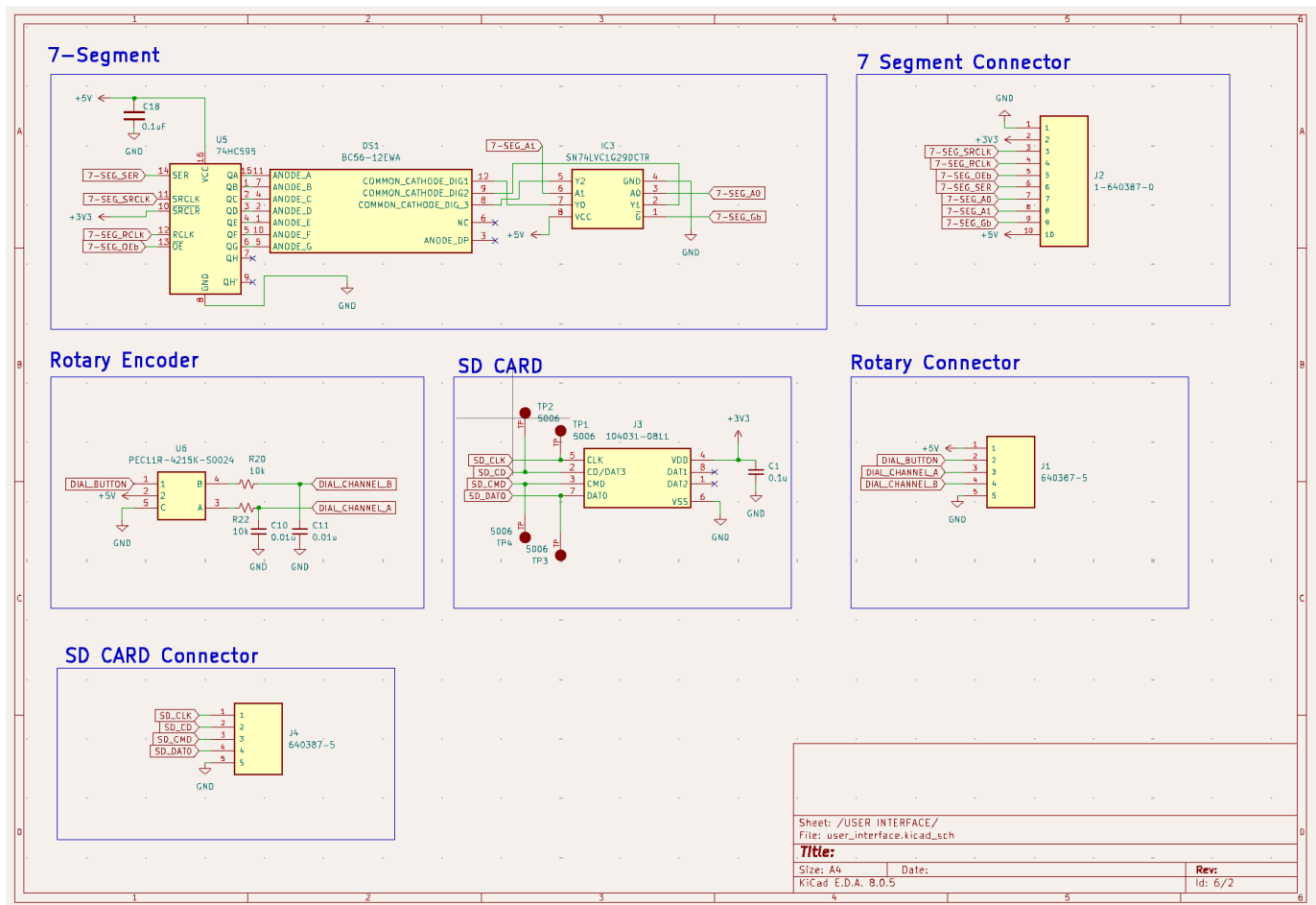
Purpose

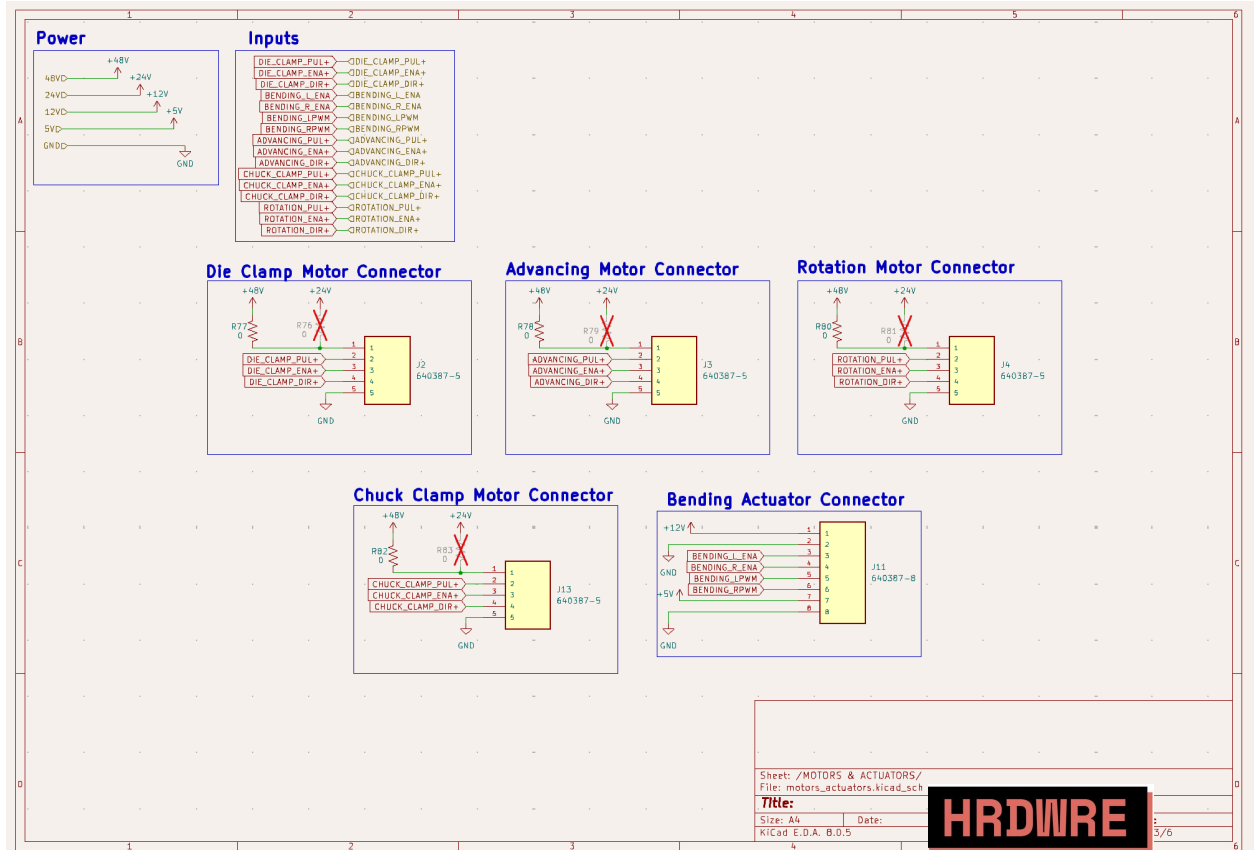
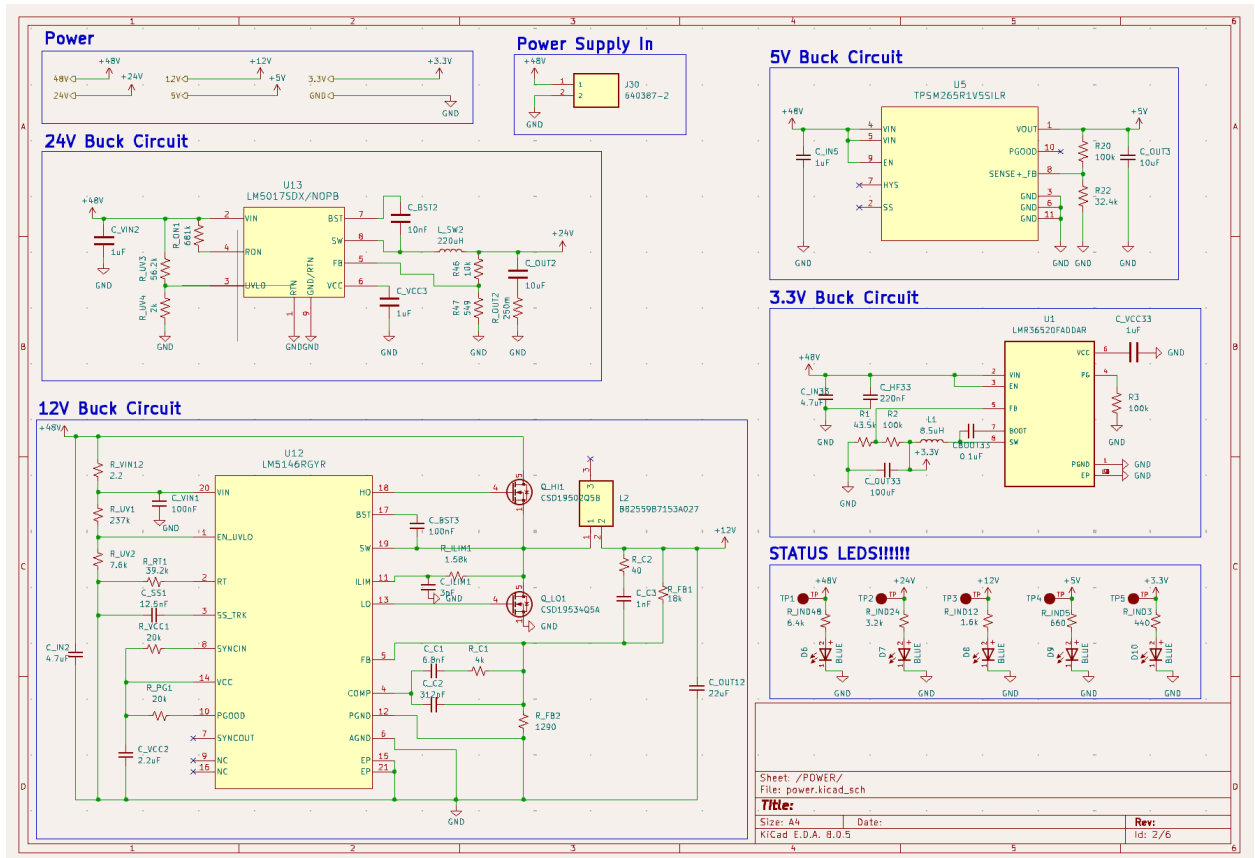
A micro SD card reader was selected for bitmap image storage, machine data logging, and retrieval of previously programmed bends.

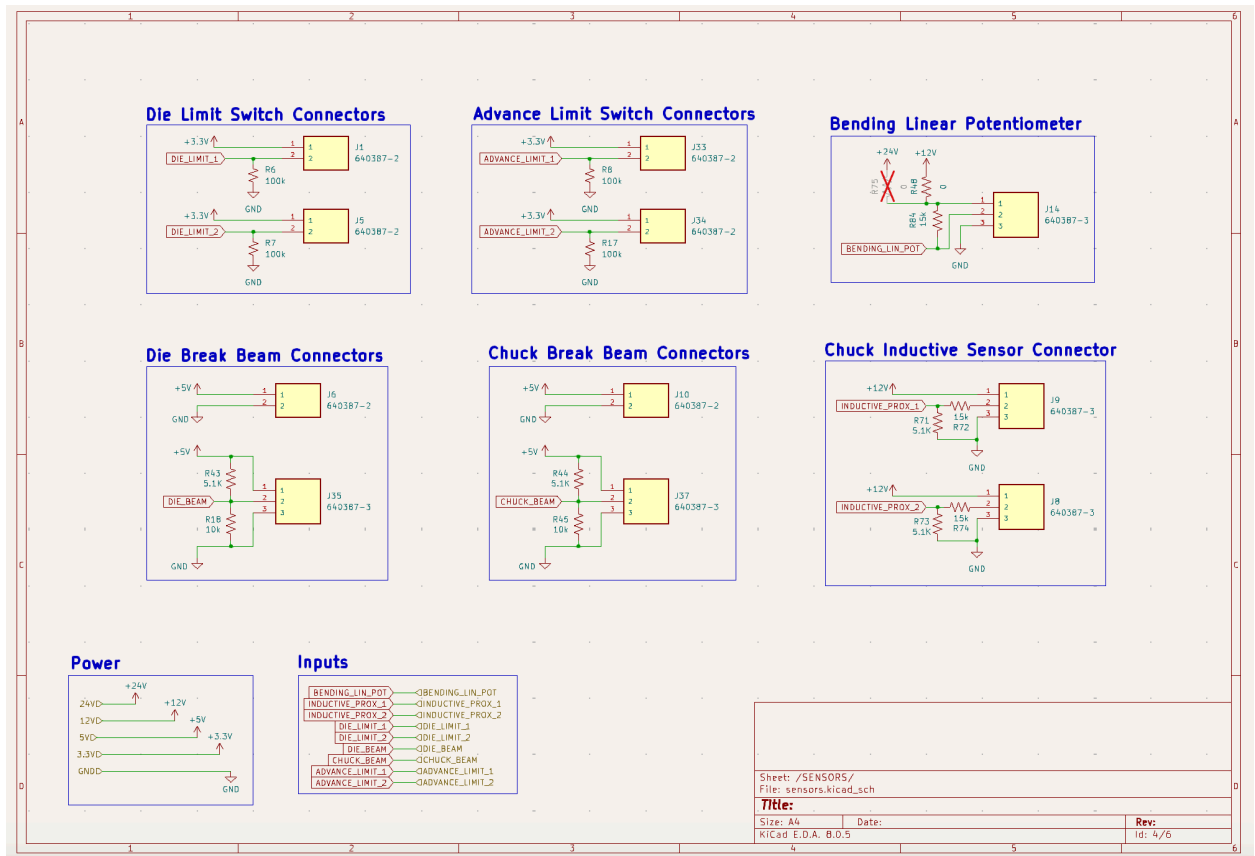
Placement and Connections

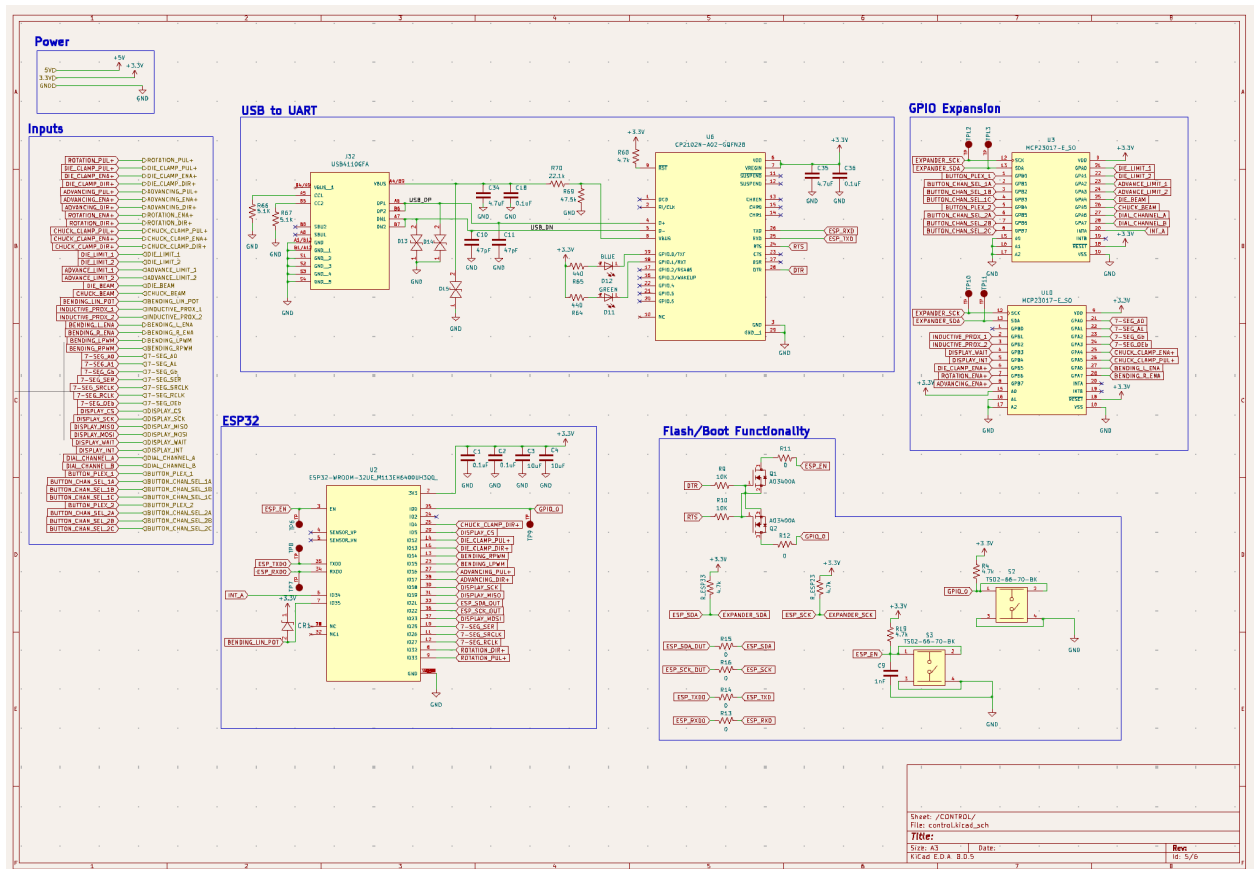
The micro SD card reader component was placed on the UI board for easier access on the front panel of the user interface. The SD card is connected to the main board via a standard connector.

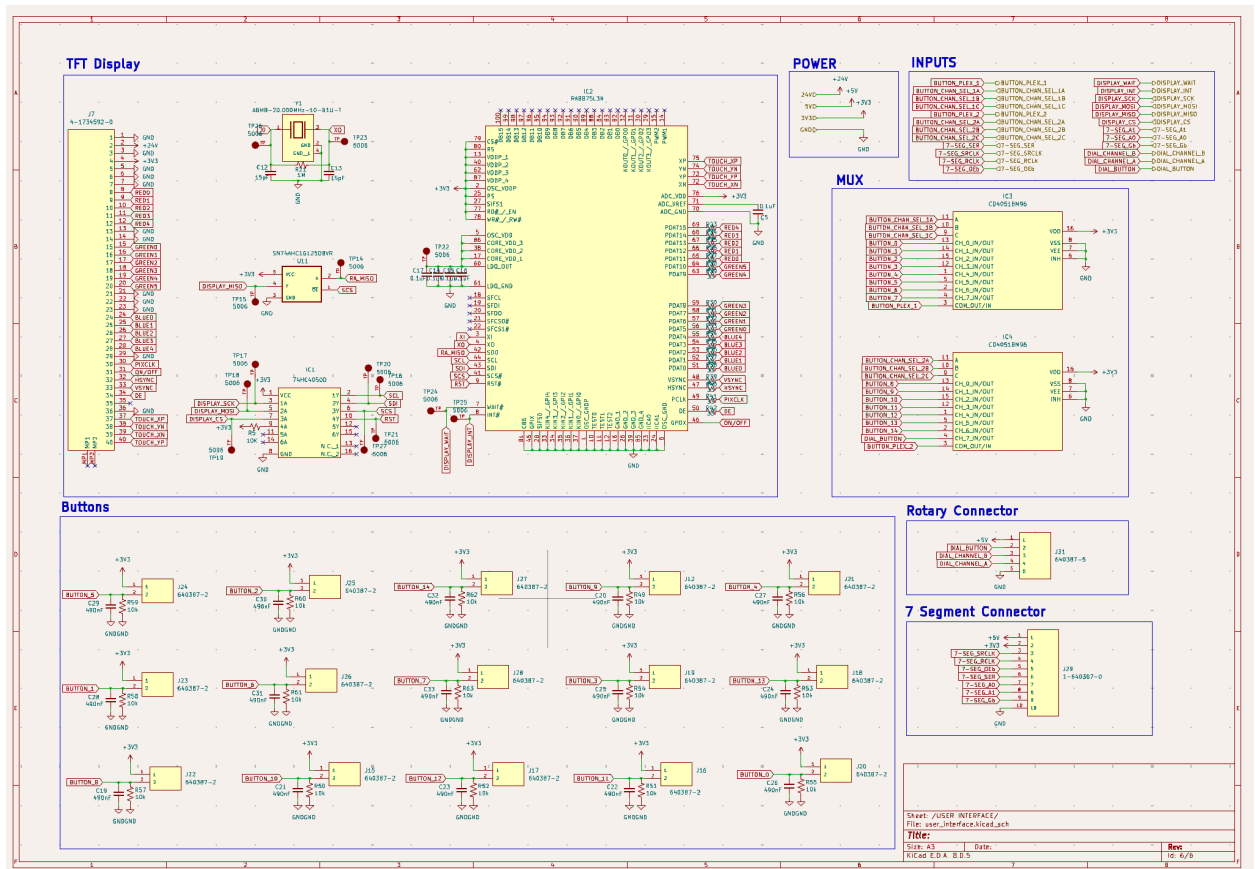
Circuit Board



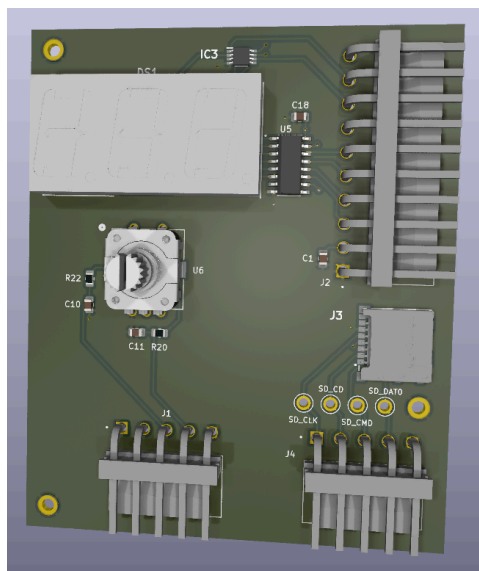


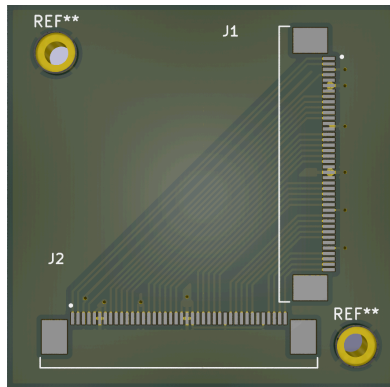






- Any input / output and where it connects to sensors, mechanical devices, etc.
- Images to show relative size and placement within the overall system





Software Design

General Software Design

- Development Environment
- Required libraries and tools
- Flowchart of software flow

- SOFTWARE DIAGRAM HERE

Mux Class:

The Mux (multiplexer) software class is designed to interact with the multiplexers that all UI buttons are attached to. The class takes input arguments of a GPIO expander object and the output pin of the multiplexer and uses these in its methods. The methods of the object allow the user to both read a specific pin of a multiplexer and scan all of the inputs of the multiplexer. When scanning, the output is an eight-bit integer, with each bit representing the corresponding position on the multiplexer.

REncoder Class:

The REncoder class is designed to read the rotary encoder. Any change to the state of the rotary encoder is determined using software interrupts, which are triggered when the rotary encoder is moved. The position of the rotary encoder can be reset using a class method, which should be done after each parameter the user inputs using the rotary encoder is stored.

The rotary encoder object contains the position of the rotary encoder and methods needed to get and set this position.

TFT Class:

The TFT class is designed to initialize, reset and write to the TFT display. The initialize method of the TFT class is run prior to any other methods and establishes communication with the TFT by beginning a SPI transaction with RA8875 graphics driver IC. After successful initialization, a total of eight methods were created for overwriting the display with specific screens as depicted in the UI screen map. Internally these display methods call the TFT class method which reset the display writing settings to defaults, and align the cursor to print text with center alignment.

PID Class:



The PID class performs PID calculations based on measurements from the linear potentiometer and the desired input bend angle, and uses these calculations to drive the linear actuator. The `prepareNewBend(angle)` method sets the setpoint for PID calculations and is intended to be called directly before the `run()` method. The method called `prepareHome()` functions the same as `prepareNewBend(angle)`, except it is for the special case of fully retracting the linear actuator. The `run()` method performs a PID calculation with feedback from the linear potentiometer every specified time interval (currently set to 50ms) and sends a new analog output value to the BTS7960 motor controller to drive the linear actuator. Once the linear actuator is within the acceptable range of values that produce the given bend angle, the `stop()` function is called, which turns off the linear actuator. For a more thorough explanation of the core PID calculations, refer to [this YouTube video](#).

Explanation of Overall Software Decisions

Communication

- Command set
 - Commands available
 - How to interpret
- Settings
 - Baud rate, necessary connections, frequency, etc.

Config Class:

The Config class is a relatively simple class for storing and validating UI inputs. It has attributes for bend types, angles, heights, lengths, tube advancements between bends, and the length of a given tube. Bends are input using the `addNewBend` method, and removed using the `removeBend` method. This class should be instantiated in the `start` method of `TubenderStateMachine` and used to track bends throughout iterations of the UI input loop. If a bend is input that decreases the remaining length of a tube beyond 0", the `addNewBend` method will return -1, indicating an error.

Testing/Version Control:

Throughout the process of software development, features were iterated using Git version control and branching. This allowed precise control over feature development, gradual integration of features as they came online, and a development process that was well integrated with testing and debugging the electrical system. Branches were used for developing different subsystems, such as UI and motors, but branches were also used to test features on the hardware. As features were tested successfully, they were integrated into the main branch.

Next Steps

- Define exactly where the project stands
 - What works? What doesn't?
 - What are the existing projects / shortcomings?
- What do you recommend doing next?
 - Getting quotes from manufacturers?
 - Gathering user feedback
 - What would you do if you had more time to work on the project?

For the main PCB, the next steps would be to revise the overall design in order to decrease the footprint of the PCB itself. While a large surface area was able to dissipate more heat than a small one, with active cooling this effect is less noticeable and less useful. However, the large amount of connectors makes it challenging to decrease the size beyond a certain point.

In the power system, the 12V buck converter should be further tested and checked for stability. While it provides the system with adequate power, simulations suggest that there may be some level of instability in the converter design.

Appendix

Appendix A1. MATLAB Code

```
% Clear all previous data and close figures
clc; clear; close all;
%% Tubender Calcs
%% Inputs
angle = 90; % bend angle [degrees]
% bending_arm = linspace(1, 10, 100); % length of die bending arm [in], array from 1 to 10 inches
theta = deg2rad(angle); % bend angle in radians
R_b = 8.25; % bending radius (die radius) [in]
mu = 0.3; % coefficient of friction for steel on steel (probably change)

%conversions
lbf_in_to_Nm = 0.113; % 1 lbf-in = 0.113 Nm
lbf_to_N = 4.44822; % 1 lbf = 4.44822 N

%% Define Variables
% Material Properties of EMT (using low carbon steel, AISI 1006 CD)
E = 10000; % Elastic modulus [ksi]
Sy = 28; % Yielding stress [ksi]
C = 1; % pinned-pinned assumption (buckling)
roh = 0.0968; % density [lb/in^3]

% Constants
g = 32.17 * 12; % gravity [in/s^2]

% Tube Dimensions
length_tube = 10 * 12; % tube length [inches]
do_tube = 1.740; % tube outer diameter [inches]
di_tube = 1.610; % tube inner diameter [inches]
ro_tube = do_tube / 2; % tube outer radius [inches]
ri_tube = di_tube / 2; % tube inner radius [inches]
V = length_tube * (pi / 4) * (do_tube^2 - di_tube^2); % tube volume [in^3]

%Dependents
weight_tube = roh * V * g; % weight of the tube
mass = roh * V; % mass of the tube [lb]
```



```
I = (pi / 64) * (do_tube^4 - di_tube^4); % Area Moment of Inertia [in^4]
A = (pi / 4) * (do_tube^2 - di_tube^2); % cross-sectional area [in^2]
J = (pi / 32) * (do_tube^4 - di_tube^4); % polar moment of inertia
Z = (pi / 32) * ((do_tube^4 - di_tube^4) / do_tube); % Section modulus
L_bend = R_b * theta; % arc length of the bent section [in]
moment_arm = 10.5; %[in]
L_lead = 13; %[in]
```

```
n = 1.25; % factor of safety
M = Sy*n*I/ro_tube*1000; % in-lb
I1 = 6; % in
I2 = 6; % in
Die_norm = M*(I1+I2)/I1/I2; % lb
F_norm = Die_norm*I2/(I1+I2); % lb
F_tot = F_norm/cosd(45) %lb
T = F_tot*I1
rpm = 2;
H = T*rpm/63000
```

```
%% Linear Acutators
%Advancing Calculations
weight_rotationassy = 30;%weight of rotational ass'y in lbs
e = 0.5; %efficiency of lead screw, lower end of 50-60%
L = 22; %total lead (22?)[in]
mu_rail = 0.003;
```

```
W = weight_tube + weight_rotationassy;
Tr = (W*0.003*L)/(2*pi*e) %[lb in]
Tr_Nm = Tr*lbf_in_to_Nm
```

```
% Moment Arm Calculations
F_arm = T/(moment_arm*cosd(45))
F_arm_N = F_arm*lbf_to_N;
```

```
%% RPM Calculation
revs = 1 * (1 / 0.1968) * (60 / (2 * pi)); % RPM
```